

Nicole Sandra-Yaffa Dumont

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Academic Appointments

Postdoctoral Researcher

Institute of Neuroinformatics (INI, UZH & ETH Zürich), Universität Zürich

📅 Dec 2025 – Present
📍 Zürich, Switzerland

- **Supervisor:** Prof. Katharina Wilmes
- Modeling neural circuits underlying uncertainty in perception and learning

Education

Ph.D in Computer Science

Diploma in Theoretical Neuroscience

University of Waterloo

📅 Sept 2019 – March 2025
📍 Waterloo, Canada

- **Supervisors:** Prof. Chris Eliasmith and Prof. Jeff Orchard
- **Dissertation:** *Symbols, Dynamics, and Maps: A Neurosymbolic Approach to Spatial Cognition*
- **Neuro-Symbolic Modeling for Spatial Representation & Mapping**
 - Developed Spatial Semantic Pointers (SSPs) as a grid cell-inspired representation of continuous features, enabling the integration of symbolic structure and uncertainty in biologically plausible neural models.
 - Built spiking neural network models of path integration, cognitive mapping, and SLAM that use neurosymbolic methods to bind continuous and discrete features to create compositional embeddings.
- **Reinforcement Learning (RL) & Action Selection**
 - Designed biologically plausible online learning rules for RL in continuous-time settings using State Space Model-based memory systems.
 - Incorporated SSPs into Bayesian Optimization and deep RL frameworks, modeling predictive inference and decision-making under uncertainty.
- **Interdisciplinary:** Research at the crossroads of cognitive science and computational neuroscience that advanced models integrating symbolic structure with neural dynamics to explain spatial cognition.

Computational Mathematics (Masters of Mathematics – Co-op Program)

University of Waterloo

📅 Sept 2017 – April 2019
📍 Waterloo, Canada

- **Supervisor:** Prof. Thomas F. Coleman
- **Thesis Project:** *The Optimal Social Cost of Carbon & The Cost of Inaction*
- Developed and analyzed a robust optimization algorithm for carbon pricing models, integrating applied mathematics methods with real-world policy modeling.

Honours Mathematics and Physics (Bachelors of Science)

McMaster University

📅 Sept 2012 – April 2017
📍 Hamilton, Canada

- Graduated with distinction. Courses on statistical mechanics, dynamical systems, cryptography, mathematical biology, quantum computing, and scientific computation.

Past Experience

Neuromorphic Scientist

Applied Brain Research

📅 Jan 2023 – Jan 2024
📍 Waterloo, Canada

- Integrated a biologically-inspired SLAM algorithm into a MuJoCo simulation environment and deployed it on low-power neuromorphic hardware (Intel Loihi 1).

Research Associate

Cayuga Research

May 2018 – Aug 2019
Waterloo, Canada

- Consulting work focused on developing optimization methods and data-driven solutions for industrial applications.
- Created prototype flight path optimization software achieving 5% fuel savings compared to real flights.
- Designed and tested chiller plant optimization models, producing 4% energy savings.

Research Assistant

Ayers Research Group, McMaster University

May 2015 – Aug 2015
Hamilton, Canada

- Implemented a semi-definite optimization algorithm constraining two-electron reduced density matrices (2-RDMs) to represent many-electron quantum systems.

Research

h-index = 6, i10-index = 4, total citations = 112 (Google Scholar)

† = These authors contributed equally

Publications & Conference Proceedings

- Furlong, P. M., Simone, K., **Dumont, N.S.**, Bartlett, M., Stewart, T. C., Orchard, J., & Eliasmith, C. (2024). Biologically-plausible markov chain monte carlo sampling from vector symbolic algebra-encoded distributions. *ICANN* [\[Link\]](#)
- Furlong, P. M., **Dumont, N.S.**, & Orchard, J. (2024). A recurrent dynamic model for efficient bayesian optimization. *NICE* [\[Link\]](#)
- **Dumont, N.S.**, Furlong, P. M., Orchard, J., & Eliasmith, C. (2023). Exploiting semantic information in a spiking neural slam system. *Frontiers in Neuroscience*, 17 [\[Link\]](#)
- **Dumont, N.S.**[†], Stöckel, A.[†], Furlong, P. M.[†], Bartlett, M., Eliasmith, C., & Stewart, T. C. (2023). Biologically-based computation: How neural details and dynamics are suited for implementing a variety of algorithms. *Brain Sciences*, 13(2) [\[Link\]](#)
- Bartlett, M.[†], Simone, K.[†], **Dumont, N.S.**[†], Eliasmith, C., & Orchard, J. Improving reinforcement learning with biologically motivated continuous state representations. In: *ICCM. 2023* [\[Link\]](#)
- **Dumont, N.S.**, Orchard, J., & Eliasmith, C. (2022). A model of path integration that connects neural and symbolic representation. *CogSci*, 44(44) [\[Link\]](#)
- Bartlett, M.[†], **Dumont, N.S.**[†], Furlong, M. P., & Stewart, T. C. Biologically-plausible memory for continuous time reinforcement learning. In: *ICCM. 2022* [\[Link\]](#)
- Voelker, A. R., Blouw, P., Choo, X., **Dumont, N.S.**, Stewart, T. C., & Eliasmith, C. (2021). Simulating and predicting dynamical systems with spatial semantic pointers. *Neural Computation*, 33(8) [\[Link\]](#)
- **Dumont, N.S.**, Stewart, T. C., & Eliasmith, C. Spiking neural network model of simultaneous localization and mapping with spatial semantic pointers. In: *CoSyNe. 2021* [\[Link\]](#)
- Coleman, T. F., **Dumont, N.S.**, Li, W., Liu, W., & Rubtsov, A. (2021). Optimal pricing of climate risk. *Computational Economics* [\[Link\]](#)
- **Dumont, N.S.** & Eliasmith, C. (2020). Accurate representation for spatial cognition using grid cells. *CogSci* [\[Link\]](#)

Manuscripts

- Furlong, P.M.[†], **Dumont, N.S.**[†], Antonova, R.[†], Orchard, J., & Eliasmith, C. (In Revision). Compositional neurosymbolic representations enable efficient active exploration. *Nature Communications*

Thesis

- **Dumont, N.S.** (2025). *Symbols, dynamics, and maps: A neurosymbolic approach to spatial cognition* [PhD Thesis]. University of Waterloo. UWSpace [\[Link\]](#)

Patents

- Voelker, A. R., Eliasmith, C. D., Blouw, P., Stewart, T., & **Dumont, N.S.** (pending, filed November 5, 2021). Methods and systems for simulating and predicting dynamical systems with vector-symbolic representations [US Patent App. 17/520,379] [\[Link\]](#)

Teaching and Supervision

- **Introduction to Computational Mathematics (2022):** Sole instructor for two sections of this course, teaching over 120 students. Planned course content and developed lectures, assignments, and exams. In the Student Course Perceptions survey, over 80% of students agreed that I helped them understand the course concepts and fostered a supportive learning environment.
- **Logic and Computation (2020):** Served as an Instructional Apprentice for an introductory course in mathematical logic. Led tutorials, graded assignments, and provided assistance during office hours.
- **Teaching Assistantships (2017-2022):** Served as a TA in six undergraduate courses in mathematics and computer science.
- **Student supervision (2020):** Supervised an undergraduate research assistant on a four-month project developing a novel algorithm for learning successor features for reinforcement learning.

Academic Service

- **Leadership & Organization**
 - Co-Organizer & Topic Area Leader, Language and Thought, Telluride Neuromorphic Cognition Engineering Workshop (2025): Successfully proposed and led a topic area, invited speakers, selected participants, gave a talk, and coordinated projects.
 - Invited Speaker & Co-Organizer, Telluride Neuromorphic Cognition Engineering Workshop (2024)
- **Instruction & Training**
 - Instructor, Nengo Summer School (2023, 2024, 2025): Taught graduate students and postdocs about neural modeling using the Nengo framework.
- **Peer Review:** Reviewer for *Nature Communications*, *Communications Biology*, *CogSci 2025*, and *Frontiers in Neuroscience*

Scholarships & Awards

- **Best New Neuromorph (2024):** Awarded to a first-time participant at Telluride who best represents the collaborative spirit of the workshop.
- **ENNS Best Paper Award (2024):** European Neural Network Society award for outstanding conference paper contribution at ICANN.
- **Barbara Hayes-Roth Award for Women in Math and Computer Science (2022)** - CAD \$1,200: Merit-based award recognizing academic excellence.
- **GO-Bell Scholarship (2019-2021)** - CAD \$24,000: Based on academic excellence and internationally competitive research potential.
- **Provost Doctoral Entrance Award for Women (2019)** - CAD \$5,000: Awarded for first-class standing and an outstanding research record.
- **Keith & Debbie Geddes Graduate Scholarship (2019)** - CAD \$5,000: Competitive internal award
- **University of Waterloo Graduate Scholarship (2018-2019)** - CAD \$3,000
- **Emanuel Williams Scholarship in Physics (2014)** - CAD \$1,200: For the second-year physics student with the highest sessional average.